MILITARY STANDARD

MIL-STD-1320-162
(NAVY)

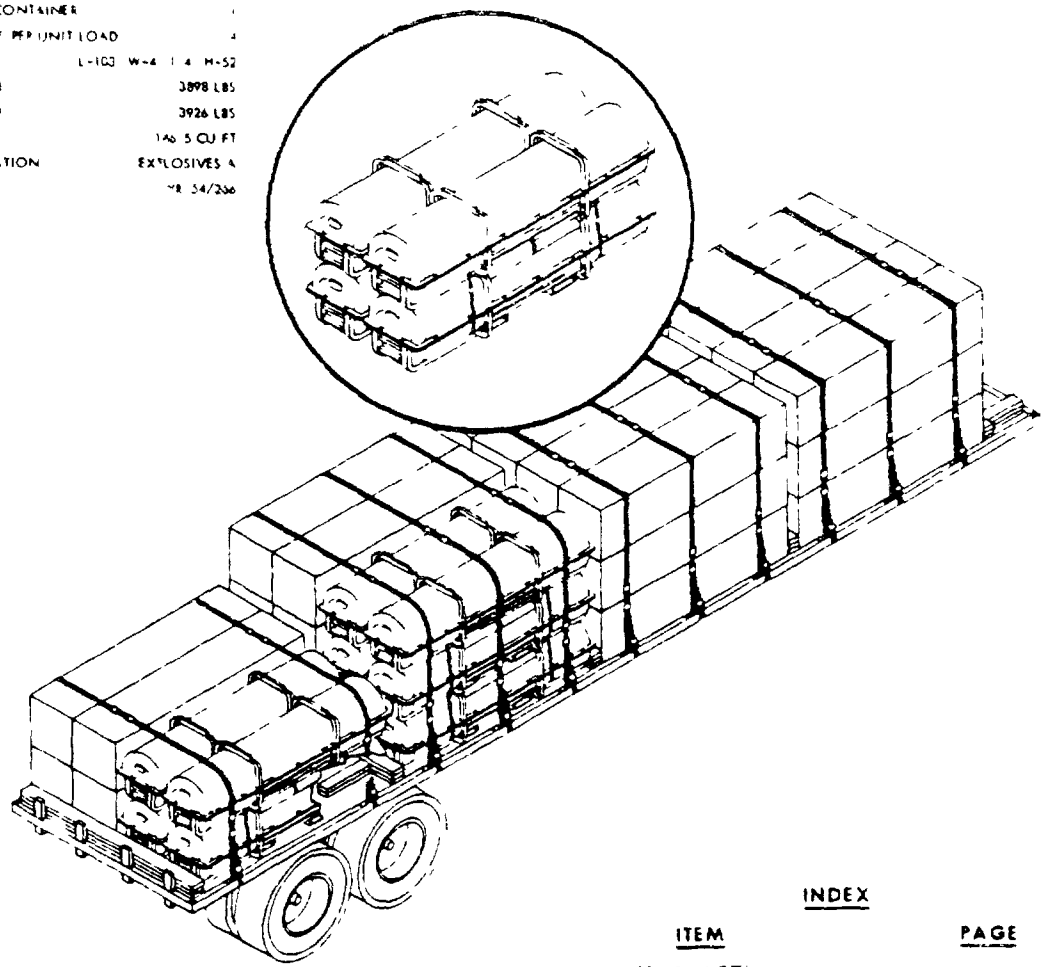
TRUCKLOADING OF HAZARDOUS MATERIALS

DISPENSER AND BOMB AIRCRAFT CBU-MK 20 AND MOD 5 (ROCKEYE II) IN CONTAINER MK 427 MOD 1 FLEET ISSUE UNIT LOAD

15 JANUARY 19[illegible]

UNIT LOAD DATA

NUMBER OF BOMBS PER CONTAINER		1
NUMBER OF CONTAINERS PER UNIT LOAD		4
DIMENSIONS		L-103 W-44 1/4 H-52
WEIGHT	DODIC E173	3898 LBS
	DODIC E819	3926 LBS
CUBE		146.5 CU FT
DOT HAZARD CLASSIFICATION		EXPLOSIVES A
UNIT LOAD DRAWING		MR 54/236



INDEX

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GENERAL NOTES	-
LOAD STRAPPING	-
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LTL 40 FT TRAILER	4 &
STACKING & STRAPPING PROCEDURE	6

NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY)

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

SIGNATURE AIRSYSCOM HQ DIRECTION DATE

ORIGINATOR [illegible]
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
NAD EARLE NEW JERSEY

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GENERAL NOTES

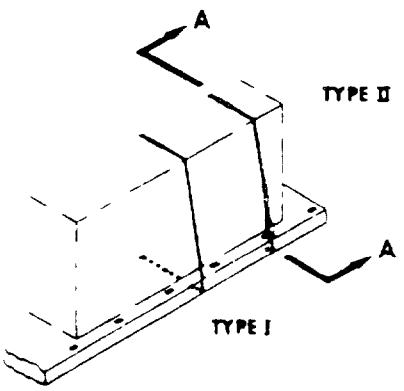
1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE AIRCRAFT DISPENSER AND BOMB CBU-[illegible]/70 AND MK D- IN CONTAINER MK 427 MOD 1. CONTAINERS SHALL BE UNITIZED FORMING THE FLEET ISSUE UNIT LOAD WR-54/266
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT FLATBED TRAILERS WITH THE TRAILER'S TANDEM AXLE LOCATED IN THE [illegible] POSITION AT THE EXTREME REAR OF THE TRAILER. DO NOT USE TRAILERS WITH METAL FLOORS WHEN REQUIRED BY THE STANDARD TWO ADDITIONAL CONTAINERS ARE STRAPPED TO A FLEET ISSUE UNIT LOAD. WHEN THE TRUCK TRANSPORTATION IS [illegible] COMPLETED THE TOP TWO CONTAINERS ARE REMOVED BY CUTTING THE TOP STRAPS AND REMOVING THE RESTRAINER. THE [illegible] CONTAINERS ARE UNITIZED PER WR-54/266 FORMING A FLEET ISSUE UNIT LOAD
4. FOR STACKING AND STRAPPING PROCEDURE SECURING THE TWO CONTAINERS TO THE FLEET ISSUE UNIT LOAD SEE PAGE 6

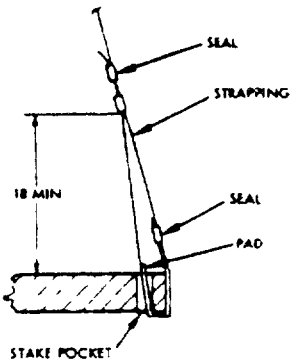
NOTE

CHAINS AND LOAD BINDERS ARE NOT AUTHORIZED FOR THIS LOAD AS THEY WILL CAUSE THE TRUCKLOAD TO BE OVERWIDTH

5. STEEL STRAPS SHALL BE [illegible] O.D AND APPLIED AS SPECIFIED BY THE LOAD STRAPPING REQUIREMENT SHOWN BELOW
6. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR TRAILER WILL NOT EXCEED THESE LIMITATIONS
7. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS TO AN ACCESSIBLE AREA AND ATTACH EXPLOSIVES "A" PLACARD TO BOTH SIDES FRONT AND BACK OF TRAILER
8. APPLICABLE MATERIAL SPECIFICATIONS DUNNAGE LUMBER MM-L-751 NAILS FF-N-105 TYPE II, STYLE 10 COMMON BRIGHT OR STRAPPING QQ-S-781 TYPE I CLASS A
9. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320

LOAD STRAPPING





SECTION A-A

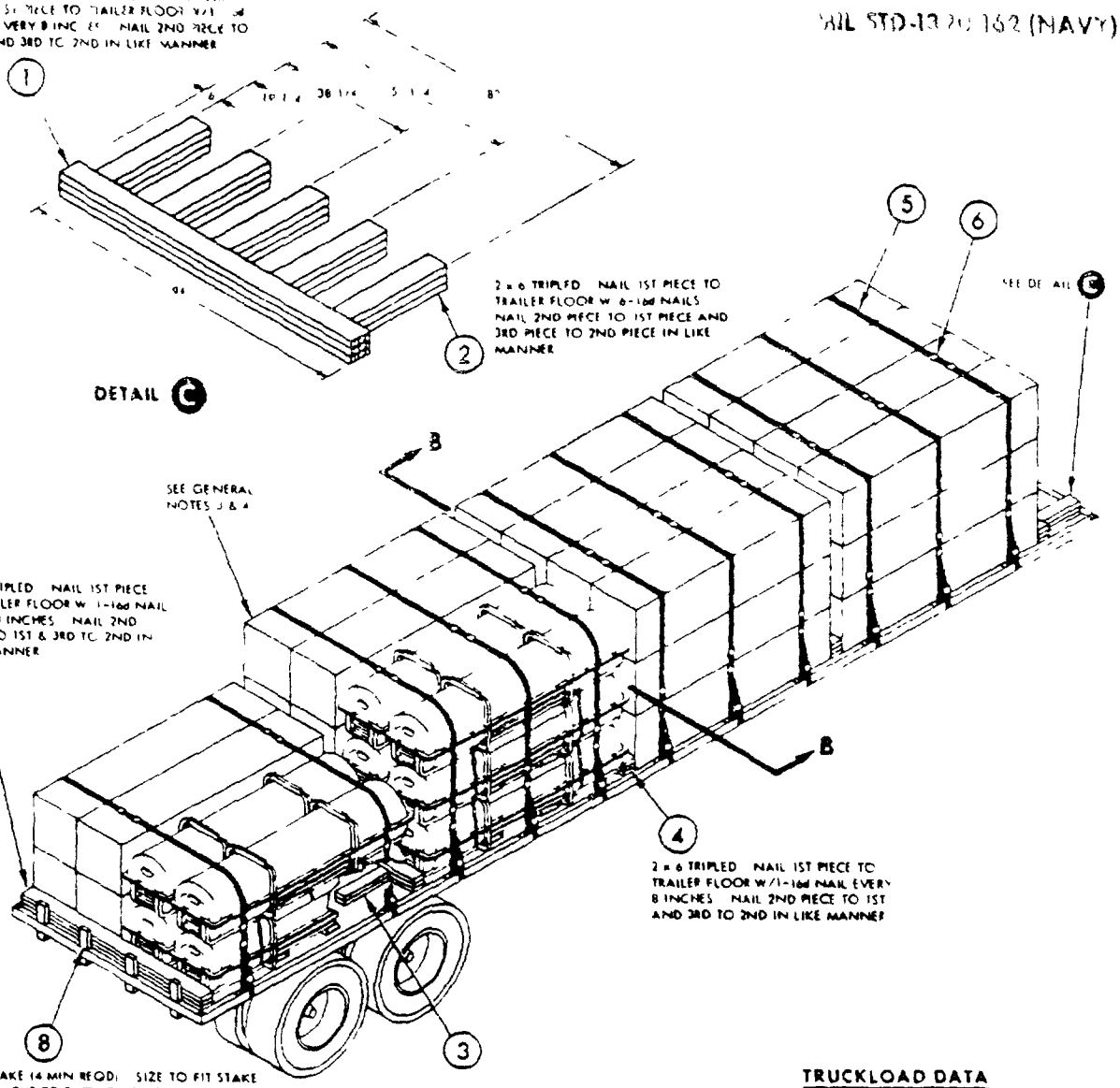
TIE DOWN STRAPPING

THE FOLLOWING TWO METHODS OF TIE-DOWN STRAPPING ARE APPROVED FOR USE WITH THE SHIPMENT OF LOADS ON FLATBED TRAILERS. STEEL STRAPPING SHALL BE IN ACCORDANCE WITH FEDERAL SPECIFICATION QQ-S-781. STRAP JOINTS SHALL HAVE A MINIMUM TENSILE STRENGTH EQUAL TO 75 PER CENT OF THE STRAP STRENGTH. FOR EACH OF THE TYPES ILLUSTRATED ABOVE IT IS PREFERRED TO POSITION TENSION AND DOUBLE CRIMP THE STRAP SEALS AT THE TOP OF THE LOAD IF PRACTICABLE.

TYPE I — CONTINUOUS STRAPPING AROUND THE FLATBED AND THE LOAD

TYPE II — THE STRAPPING IS SECURED TO THE STAKE POCKETS ONE PIECE ON EACH SIDE OF THE TRAILER AND IS BROUGHT UP OVER THE LOAD TENSIONED AND SEALED WITH TWO DOUBLE-CRIMPED SEALS ON THE TOP. METHOD OF SECURING STRAPPING TO STAKE POCKET IS SHOWN IN SECTION A-A. THE SHORT END IS ON THE INSIDE AND IS SECURED WITH TWO DOUBLE CRIMPED SEALS AT A MINIMUM OF 18 INCHES ABOVE THE TRAILER BED. A STAKE POCKET PAD, A SHORT PIECE OF THE SAME STRAPPING 18 INCHES LONG IS INSERTED BETWEEN THE MAIN STRAP AND THE STAKE POCKET AND IS SECURED TO THE MAIN STRAP WITH A SEAL AS SHOWN.

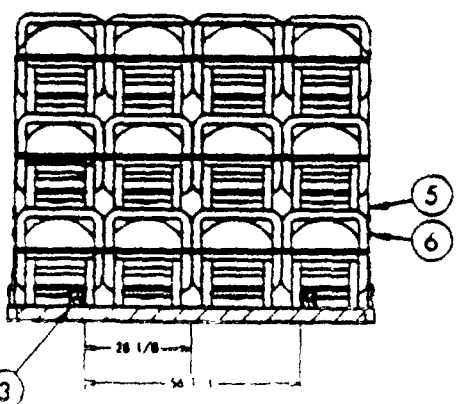
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STUD STAKE (4 MIN REQD) SIZE TO FIT STAKE POCKET CUT TO EXTEND APPROX 9" ABOVE TRAILER FLOOR AND 3" BELOW STAKE POCKET DRIVE 1 20d NAIL INTO STAKE BELOW STAKE POCKET ALLOWING APPROX 1" TO PROTRUDE TO PREVENT STAKE FROM WORKING OUT OF PLACE

TRUCKLOAD DATA

NUMBER OF UNIT LOADS		11
WEIGHT (APPROX)	DODIC E173	42 878 LBS
	DODIC EB19	43 186 LBS



2 x 4 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR W/ 6 20d NAILS NAIL 2ND PIECE TO 1ST IN LIKE MANNER

SECTION B-B

PIECE NO	DESCRIPTION	SIZE	NO PCS REQT	NAIL TO	NUMBER NAILS	SIZE
8	STUB STAKE	TO SUIT	4 MIN	SEE FIELD NOTE		
7	AFT CROSSMEMBER	2 x 6 x 92	3	SEE FIELD NOTE		
6	SEAL	FOR 2 IN STRAP	88			
5	STRAP	2 x 035 LENGTH TO SUIT	11	-		-
4	SEPARATOR	2 x 6 x 9[illegible]	9	SEE FIELD NOTE		
3	SLEEPER	2 x 4 x 24	32	SEE SECTION B-B		
2	CLEAT	1 x 6 x 30	[illegible]	SEE DETAIL C		
1	FWD CROSSMEMBER	2 x 6 x 92		SEE DETAIL C		

LIST OF MATERIALS AND NAILING DATA

FIL 40 FT TRAILER (FLATBED)

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LTL 40 FT TRAILER (FLATBED)

1 WHEN REQUIRED TO SHIP LESS THAN-TRUCKLOAD SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER OF UNIT LOADS TO BE SHIPPED FROM TABLE I AND POSITION THE UNIT LOADS ON THE TRAILER ACCORDINGLY

2 WHEN LOADING 8 9 OR 10 UNIT LOADS BLOCK AND BRACE AS SHOWN ON PAGE 2 ADJUSTING THE NUMBER OF UNIT LOADS AS REQUIRED

NOTE

STACKS 3 CONTAINERS HIGH REQUIRE 3 TIE DOWN STRAPS ALL OTHER STACKS REQUIRE 2 TIE-DOWNS

3 WHEN LOADING 7 UNIT LOADS BLOCK AND BRACE AS SHOWN ON PAGE 5

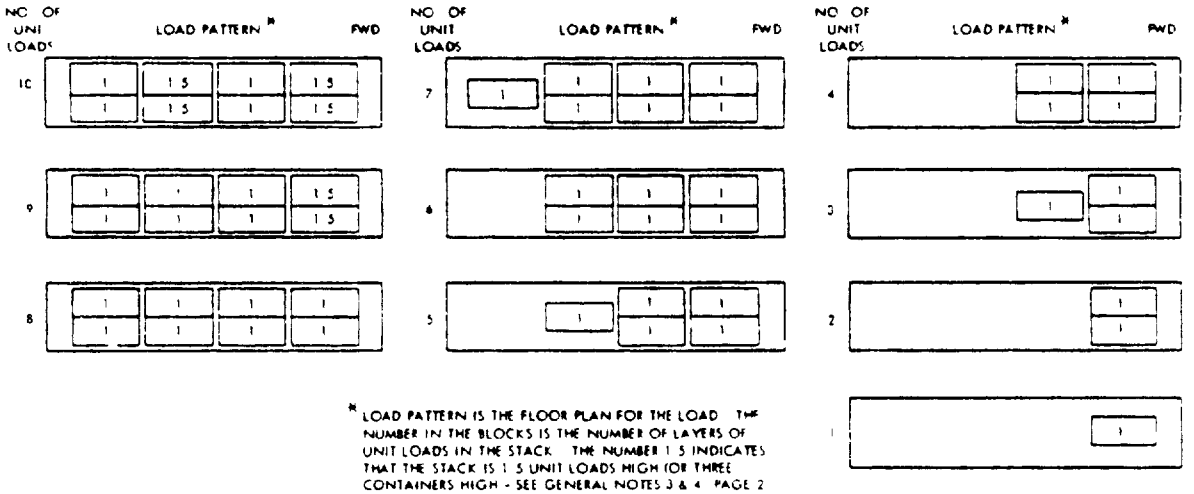
4 WHEN LOADING 6 OR LESS UNIT LOADS BLOCK AND BRACE USING THE PRINCIPLES SHOWN ON PAGE 5 USE DETAIL C PAGE 3 OR DETAIL D PAGE 4 AT THE REAR OF THE LOAD AS APPROPRIATE

5 WHEN THE LOAD PLAN REQUIRES THE STACK TO BE 1 5 UNIT LOADS HIGH (THREE CONTAINERS HIGH THEY SHALL BE STRAPPED TOGETHER AS SHOWN ON PAGE 6 BLOCK BRACE AND TIE-DOWN AS SPECIFIED ON PAGE 2

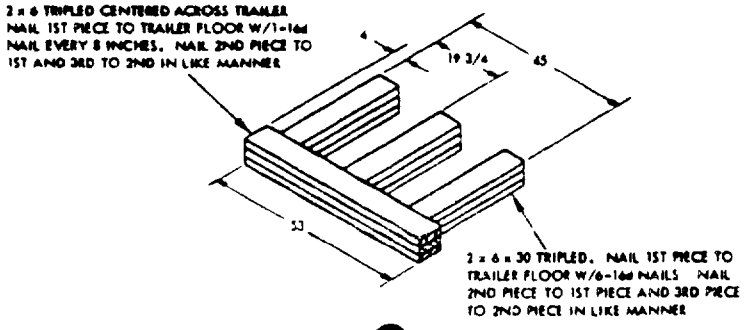
6 THE LTL SHOWN ON PAGE 5 IS FOR SEVEN UNIT LOADS AND SHOWS THE CORRECT BLOCKING BRACING AND USE OF TIE-DOWNS FOR A LOAD OF THIS QUANTITY

7 ACTUAL SHIPPING REQUIREMENTS WILL DETERMINE THE NUMBER OF UNIT LOADS TO BE SHIPPED THE LOAD PATTERN BLOCKING BRACING AND TIE-DOWNS SHALL BE IN ACCORDANCE WITH THE APPROPRIATE PARTS OF THIS MILITARY STANDARD AND THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY)

TABLE I



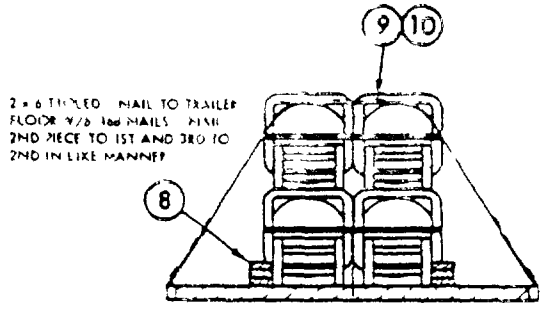
* LOAD PATTERN IS THE FLOOR PLAN FOR THE LOAD THE NUMBER IN THE BLOCKS IS THE NUMBER OF LAYERS OF UNIT LOADS IN THE STACK THE NUMBER 1 5 INDICATES THAT THE STACK IS 1 5 UNIT LOADS HIGH (OR THREE CONTAINERS HIGH - SEE GENERAL NOTES 3 & 4 PAGE 2



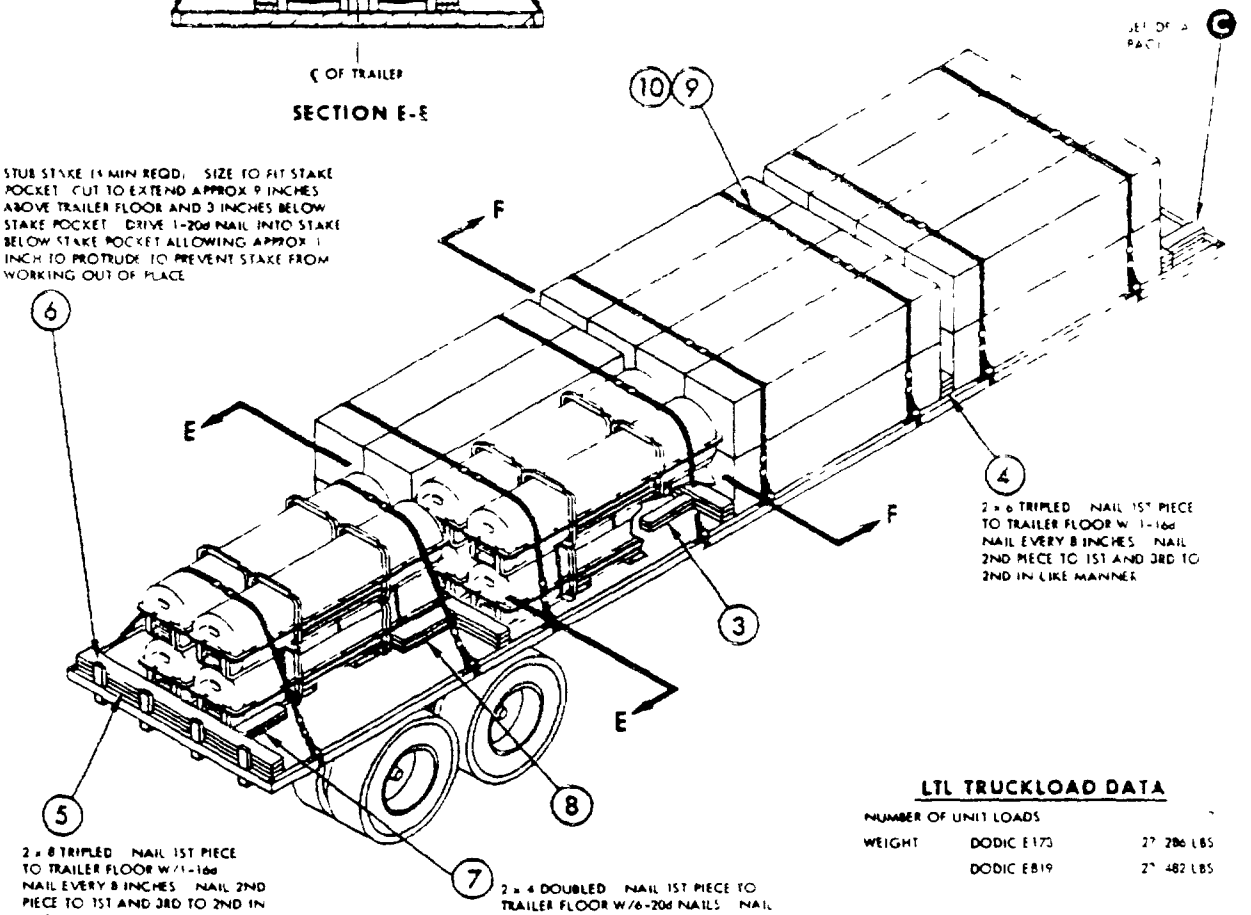
DETAIL D

FRONT AND REAR BLOCKING
(SEE LTL NOTE 4)

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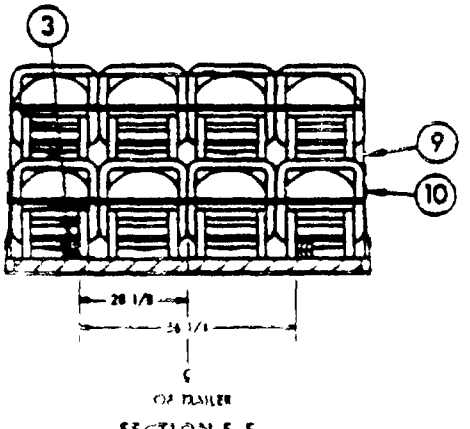
SECTION E-E



LTL TRUCKLOAD DATA

NUMBER OF UNIT LOADS		
WEIGHT	DODIC E173	27 286 LBS
	DODIC E819	27 482 LBS

2 x 4 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR W/6-20d NAILS NAIL 2ND PIECE TO 1ST IN LIKE MANNER



SECTION F-F

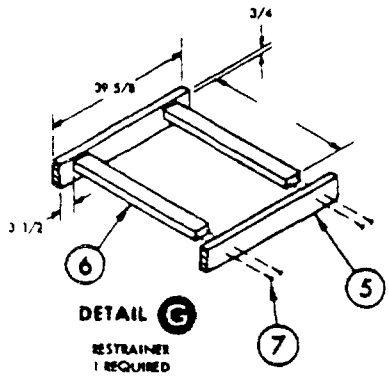
PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NO PER NAILS	SIZE
10	SEAL	FOR 2 IN STRAP	64			
9	STRAP	2 x .050 LENGTH TO SUIT	8	-		-
8	BACK UP	2 x 6 x 20	6	SEE SECTION E-E		
7	SLEEPER	2 x 4 x 24	4	SEE FIELD NOTE		
6	STUB STAKE	TO SUIT	4 MIN	SEE FIELD NOTE		
5	AFT CROSSMEMBER	2 x 8 x 92	3	SEE FIELD NOTE		
4	SEPARATOR	2 x 6 x 92	9	SEE FIELD NOTE		
3	SLEEPER	2 x 4 x 24	24	SEE SECTION F-F		
2	CLEAT	4 x 6 x 30	15	SEE DETAIL C PAGE 3		
1	FWD CROSSMEMBER	2 x 6 x 92	3	SEE DETAIL C PAGE 3		

LIST OF MATERIALS AND NAILING DATA

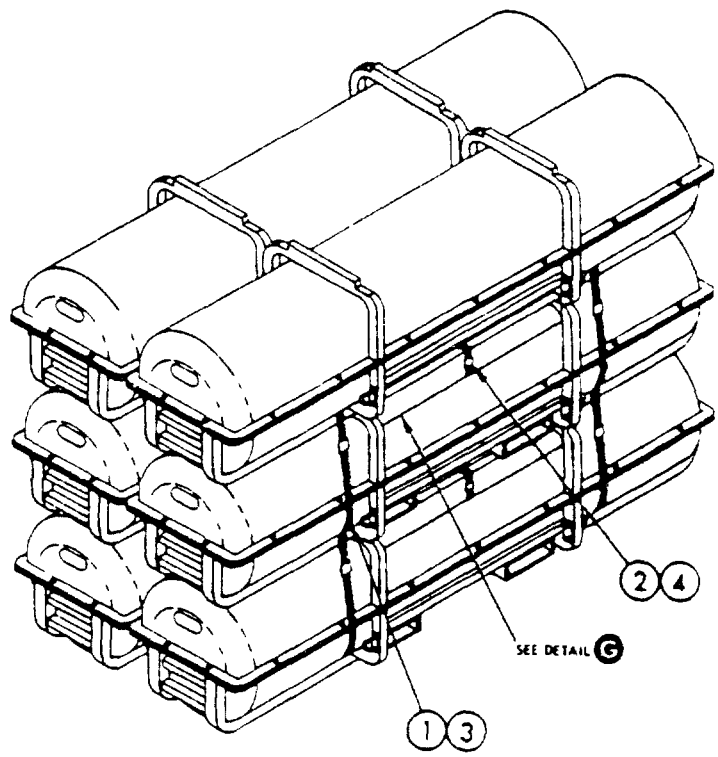
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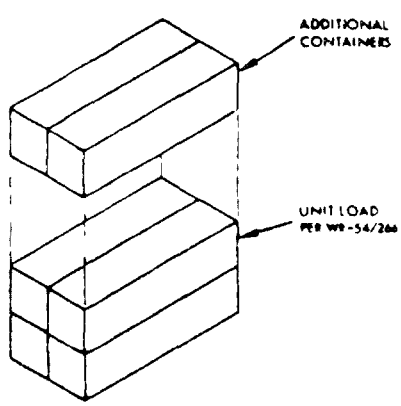
STACKING AND STRAPPING PROCEDURE

1. POSITION TWO (2) CONTAINERS ON TOP OF UNIT LOAD OF CONTAINER USING CARE TO ALIGN STACKING BAND. (SEE DETAIL H AND MAIN VIEW
   POSITION STRAPPING (ITEM 1) UNDER THE BODY OF THE CENTER LAYER OF CONTAINERS AND OVER SKIDS OF TOP TWO CONTAINERS TENSION AND DOUBLE CRIMP SEAL (ITEM 3)
3. PARTIALLY ASSEMBLE RESTRAINER (DETAIL G) AS SHOWN. INSERT RESTRAINER SPACERS THROUGH HOLES (LIFT POCKETS) IN TOP TWO CONTAINERS. ASSEMBLE REMAINING PIECE (ITEM 5). STRAP ITEM 5 TOGETHER WITH STRAPPING (ITEM 2) THREADING STRAP BETWEEN LAYERS OF CONTAINERS. TENSION STRAPPING AND DOUBLE CRIMP SEAL (ITEM 4



DETAIL G
RESTRAINER
1 REQUIRED





DETAIL H

* 4130 COLD ROLLED HEAT TREATED MAY BE SUBSTITUTED

REQ	ITEM	DESCRIPTION	MAT'L	MAT'L SPEC.	DIMENSIONS
8		NAIL	STEEL CEMENT CTD	FF-N-105	10d
2	6	RESTRAINER SPACER	WOOD DRESSED	MIL-STD-731	2 x 4 x 39 1/8
2	5	RESTRAINER	WOOD DRESSED	MIL-STD-731	2 x 6 x 39 5/8
1	4	SEAL	STEEL ZINC CTD	COM L	3/4 x 22 GA MIN
2	3	SEAL	STEEL ZINC CTD	COM L	1 1/4 x 22 GA MIN
1	2	STRAPPING	STEEL ZINC CTD	QQ-S-781 TY I CL B	3/4 x .035 x 9 FT 6
2	1	STRAPPING	STEEL ZINC CTD	QQ-S-781 TY I CL B	1 1/4 x .035 x 12 FT 6

LIST OF MATERIALS

REVIEW ACTIVITY
NAVY - CL A

PREPARING ACTIVITY
NAVY - OS
(PROJECT NO. B140-N 267

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